

Test report No:

6213330.50

TEST REPORT

Electromagnetic Compatibility (EMC)

Identification of item tested	Charging stand
Trademark	N/A
Model and /or type reference	232B, J2B, JZ01, ET9, ET12, ET19, ET10, ET20
Ratings	Class II Input: 220-240 Vac, 50-60 Hz, 600 W, 0,8 A (for charging only) Output: 27,0 Vdc, 0,8 A
Test Laboratory / address	DEKRA Testing and Certification (Shanghai) Ltd. No.250, Jiangchangsan Road, Jing'an District, Shanghai, China
Applicant's name / address	Jiangsu CHO Technology Co., Ltd 299 Xinyang Road, Ehu Town, Xishan District, Wuxi City Jiangsu, China
Test method requested, standard	EN IEC 55014-1:2021 EN IEC 55014-2:2021 EN IEC 61000-3-2:2019+A1:2021+A2:2024 EN 61000-3-3:2013+A1:2019+A2:2021
Verdict Summary	IN COMPLIANCE
Tested by (name / position & signature)	Vincent Gao Project Manager 
Approved by (name / position & signature)	Stefan Zhao Senior Project Manager 
Date of issue	2025-03-26
Report template No	TRF_EN55014-1_EN55014-2_EMC01 V1.1

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COMPETENCES AND GUARANTEES

DEKRA is a testing laboratory competent to carry out the tests described in this report.

In order to assure the traceability to other national and international laboratories, DEKRA has a calibration and maintenance program for its measurement equipment.

DEKRA guarantees the reliability of the data presented in this report, which is the result of the measurements and the tests performed to the item under test on the date and under the conditions stated in the report and it is based on the knowledge and technical facilities available at DEKRA at the time of performance of the test.

DEKRA is liable to the client for the maintenance of the confidentiality of all information related to the item under test and the results of the test.

The results presented in this Test Report apply only to the particular item under test established in this document.

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GENERAL CONDITIONS

1. This report is only referred to the item that has undergone the test.
2. This report does not constitute or imply on its own an approval of the product by the Certification Bodies or Competent Authorities.
3. This document is only valid if complete; no partial reproduction can be made without previous written permission of DEKRA.
4. This test report cannot be used partially or in full for publicity and/or promotional purposes without previous written permission of DEKRA.

UNCERTAINTY

For all measurements where guidance for the calculation of the instrumentation uncertainty of a measurement is specified in EN 55016-4-2 (CISPR 16-4-2), EN/IEC 61000-4 series or a product standard, the measurement instrumentation uncertainty has been calculated and applied in accordance with these standards.

Uncertainties have been calculated according to the DEKRA internal document PROD-P-EMC-M22. The reported expanded uncertainties are based on a standard uncertainty multiplied by a coverage factor of $k=2$, providing a level of confidence of approximately 95%. Refer to the Annex 1 for further information.

The measurement result is considered in conformance with the requirement if it is within the prescribed limit, It is not necessary to calculate the uncertainty associated with the measurement result.

ENVIRONMENTAL CONDITIONS

The climatic conditions during the tests are within the limits specified by the manufacturer for the operation of the EUT and the test equipment. The climatic conditions during the tests were within the following limits:

Ambient temperature	15 °C – 35 °C
Relative Humidity air	30% - 60%
Atmospheric pressure	86 kPa – 106 kPa

If explicitly required in the basic standard or applied product / product family standard the climatic values are recorded and documented separately in this test report.

POSSIBLE TEST CASE VERDICTS

Test case does not apply to test object	N/A
Test object does meet requirement	P (Pass) / PASS
Test object does not meet requirement	F (Fail) / FAIL
Not measured	N/M

DEFINITION OF SYMBOLS USED IN THIS TEST REPORT

☒	Indicates that the listed condition, standard or equipment is applicable for this report/test/EUT.		
☐	Indicates that the listed condition, standard or equipment is not applicable for this report/test/EUT.		
Decimal separator used in this report	☒	Comma (,)	☐ Point (.)

ABBREVIATIONS

For the purposes of the present document, the following abbreviations apply:

EUT	: Equipment Under Test
QP	: Quasi-Peak
CAV	: CISPR Average
AV	: Average
CDN	: Coupling Decoupling Network
SAC	: Semi-Anechoic Chamber
OATS	: Open Area Test Site
BW	: Bandwidth
AM	: Amplitude Modulation
PM	: Pulse Modulation
HCP	: Horizontal Coupling Plane
VCP	: Vertical Coupling Plane
U_N	: Nominal voltage

DATA PROVIDED BY THE APPLICANT

The following data has been provided by the client:

1. Information relating to the description of the sample ("Identification of the item tested", "Trademark", "Model and/or type reference tested", "Applicant's name / address", "Ratings" and "Derived model(s)").
2. Derived model not tested. These models have been declared by the applicant as being the same as the model under test.

DEKRA declines any responsibility with respect to the information provided by the applicant and that may affect the validity of results.

DOCUMENT HISTORY

Report nr.	Date	Description
6213330.50	2025-03-26	First release.

REMARKS AND COMMENTS

The equipment under test (EUT) does meet the essential requirements of the stated standard(s)/test(s).

The test results relate only to the samples tested.

J2B, JZ01, ET9, ET12, ET19, ET10, ET20 are same as 232B except for model name.

Therefore, 232B was selected for the full tests and the results are also representative for the others.

1 GENERAL INFORMATION

1.1 General Description of the Item(s)

Description of the item	Charging stand
Test model / Type number	232B
Serial number	N/A
Trademark.....	N/A
Manufacturer.....	Jiangsu CHO Technology Co., Ltd 299 Xinyang Road, Ehu Town, Xishan District, Wuxi City Jiangsu, China
Factory	Jiangsu CHO Technology Co., Ltd 299 Xinyang Road, Ehu Town, Xishan District, Wuxi City Jiangsu, China

Rated power supply	Voltage and Frequency		Reference poles				
			L1	L2	L3	N	PE
	☐	AC: 100 – 240 V, 50/60 Hz	☐	☐	☐	☐	☐
	☒	AC: 220 – 240 V, 50-60 Hz	☒	☐	☐	☒	☐
	☐	DC: 12 V, 24 V, 12 / 24 V					
	☐	Battery:					
Rated Power	600 W						
Clock frequencies	<15 MHz						
Other parameters.....	Class II						
Software version	N/A						
Hardware version.....	N/A						
Mounting position.....	☐	Table top equipment					
	☐	Wall/Ceiling mounted equipment					
	☒	Floor standing equipment					
	☐	Hand-held equipment					
	☐	Other:					

Intended use of the Equipment Under Test (EUT)
The apparatus as supplied for the test is a Charging stand, intended for residential and indoor use. These products have electronic control unit.

No	Module/parts of test item	Type	Manufacturer
	N/A		

No	Documents as provided by the applicant - Description	File name	Issue date
	N/A		

Modifications to the test item during testing	☒	N/A	☐	Refer to the chapter 6.
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Copy of marking plate:
N/A

1.2 Environment

The requirements and standards apply to equipment intended for use in:

☒	Residential (domestic) environment.
☐	Commercial and light-industrial environment.
☐	Industrial environment.

1.3 Test data

Test Location	DEKRA Testing and Certification (Shanghai) Ltd. No.250, Jiangchangsan Road, Jing'an District, Shanghai, China
Date (receive sample)	2024-12-20 (sample(s) provided by applicant)
Date (start test)	2024-12-20
Date (finish test)	2024-12-30

1.4 Classification according to EN IEC 55014-2

The standard EN IEC 55014-2 is subdivided in five categories. For each category, specific immunity requirements are formulated.

☐	Category I: equipment containing no electronic control circuitry. <u>Examples:</u> Appliances, tools and toys that contain no electronic control circuits and only electromechanical components such as switches, thermostats, brush motors, induction motors, heating elements, lighting toys containing only batteries and LED. Electric circuits consisting of passive components (such as radio interference suppression capacitors or inductors, mains transformers and mains frequency rectifiers) are not considered to be electronic control circuitry.
☒	Category II: mains operated equipment containing electronic control circuitry with no clock frequency higher than 15 MHz.
☐	Category III: battery operated equipment not included in Category I. <u>Examples:</u> Appliances, tools and toys powered by batteries and that include a microprocessor to provide a selection of functions. NOTE The assignment to Category III is independent of the clock frequency. This category also includes equipment provided with rechargeable batteries, which can be charged, directly or indirectly, from the mains. Accordingly, this equipment shall also be subjected to the test requirements for mains operated equipment but only when testing the charging function. If the equipment can operate its intended functions when connected, directly or indirectly to the mains, then it is not battery operated. Accordingly, it shall be classified as Category II, Category IV or Category V, as applicable, and subjected to the corresponding test requirements when in mains operation.
☐	Category IV: mains operated equipment containing electronic control circuitry with a highest clock frequency greater than 15 MHz but lower than or equal to 200 MHz.
☐	Category V: mains operated equipment containing electronic control circuitry with a highest clock frequency greater than 200 MHz.
Clock frequency: Fundamental frequency of any signal used in the device, excluding those which are solely used inside integrated circuits (IC).	

2 DESCRIPTION OF TEST SETUP

2.1 Operating mode(s) used for tests

During the tests the following operating mode(s) has(have) been used.

Operating mode	Operating mode description	Used for testing	
		Emission	Immunity
1	Charge mode	☒	☐
2	Dust collection mode	☒	☒
3		☐	☐
<u>Supplemental information:</u>			

2.2 Port(s) of the EUT

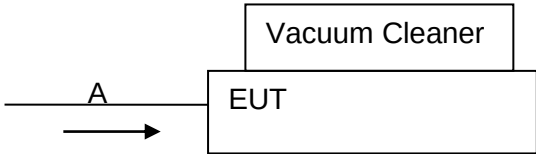
Port name and description	Connected to / Termination	Cable		
		Length used during test [m]	Attached during test	Shielded
AC input power port	AC mains	1,8	☒	☐
			☐	☐
			☐	☐
			☐	☐
			☐	☐
<u>Supplemental information:</u>				

2.3 Support / Auxiliary equipment / unit / software for the EUT

The EUT has been tested with the following auxiliary equipment / unit / software:

Auxiliary equipment / unit / software	Type / Version	Manufacturer	Supplied by
Vacuum Cleaner ¹⁾	232	Jiangsu CHO Technology Co., Ltd	Applicant
<u>Supplemental information:</u>			
1) Vacuum Cleaner is not part of EUT.			

2.4 **Test Configuration / Block diagram used for tests**

Test Configuration / Block diagram		
		
Cable Type		Signal cable Description
A	AC input power port	Non-Shielded, 1,8 m, 2pin

3 VERDICT SUMMARY SECTION

This chapter presents an overview of standards and results. Refer to the next chapters for details of measured test results and applied test levels.

3.1 Standards

Standard	Year	Description
EN IEC 55014-1	2021 ¹⁾	Requirements for household appliances, electric tools and similar apparatus – Part 1: Emission.
EN 55016-2-1 +A1	2014 2017	Methods of measurement of disturbances and immunity - Conducted disturbance measurements.
EN 55016-2-2	2011	Methods of measurement of disturbances and immunity – Measurement of disturbance power.
EN 55016-2-3 +A1	2017 2019	Methods of measurement of disturbances and immunity - Radiated disturbance measurements.
EN IEC 61000-3-2 +A1 +A2	2019 ¹⁾ 2021 ¹⁾ 2024 ¹⁾	Limits for harmonic current emissions (equipment input current ≤ 16 A per phase).
EN 61000-3-3 +A1 +A2	2013 2019 ¹⁾ 2021 ¹⁾	Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection.
EN IEC 55014-2	2021 ¹⁾	Requirements for household appliances, electric tools and similar apparatus – Part 2: Immunity – Product family standard.
EN 61000-4-2	2009	Electrostatic discharge immunity test.
EN 61000-4-3 +A1 +A2	2006 2008 2010	Radiated, radio-frequency, electromagnetic field immunity test.
EN 61000-4-4	2012	Electrical fast transient/burst immunity test.
EN 61000-4-5 +A1	2014 2017	Surge immunity test.
EN 61000-4-6	2014	Immunity to conducted disturbances, induced by radio-frequency fields.
EN IEC 61000-4-11	2020	Voltage dips, short interruptions and voltage variations immunity tests.

¹⁾ Not harmonized yet.

3.2 Deviation(s) from the Standard(s) / Test Specification(s)

The following deviation(s) was / were made from the published requirements of the listed standards:
 N/A.

3.3 Overview of results

EMISSION TESTS – EN IEC 55014-1			
Requirement – Test case	Basic standard(s)	Verdict	Remark
Conducted disturbance at mains terminals (150 kHz – 30 MHz)	EN 55016-2-1	PASS	---
Conducted disturbance at wired network ports (150 kHz – 30 MHz)	EN 55016-2-1	N/A	See 1)
Disturbance power (30 MHz – 300 MHz)	EN 55016-2-2	PASS	---
Radiated disturbance (30 MHz – 1000 MHz)	EN 55016-2-3	N/A	See 3)
Radiated disturbance (1 GHz – 6 GHz)	EN 55016-2-3	N/A	See 2)
Discontinuous disturbance (clicks) on AC power leads	EN IEC 55014-1	PASS	---
<u>Supplementary information:</u> 1) The test is not applicable as the EUT does not have associated ports / wired network ports. 2) The highest internal frequency of the EUT is less than 108 MHz. 3) According to clause 4.3.4.2 procedure (a) of the CISRP 14-1 standard the EUT is deemed to comply in the frequency range from 300 MHz to 1000 MHz without further measurements.			

EMISSION TESTS – EN IEC 61000-3-2, EN 61000-3-3			
Requirement – Test case	Basic standard(s)	Verdict	Remark
Harmonic current emissions	EN IEC 61000-3-2	PASS	---
Voltage changes, voltage fluctuations and flicker	EN 61000-3-3	PASS	---
<u>Supplementary information:</u>			

IMMUNITY TESTS – EN IEC 55014-2			
Requirement – Test case	Basic standard(s)	Verdict	Remark
Electrostatic discharge	EN 61000-4-2	PASS	---
Radio-frequency electromagnetic fields	EN 61000-4-3	N/A	See 1)
Fast transients	EN 61000-4-4	PASS	---
Surges	EN 61000-4-5	PASS	---
Injected currents (radio-frequency common mode)	EN 61000-4-6	PASS	---
Voltage dips and short interruptions	EN IEC 61000-4-11	PASS	---
<u>Supplementary information:</u> 1) The equipment is classified as category II equipment according to EN IEC 55014-2, no radio-frequency electromagnetic fields immunity test is applicable.			

4 EMISSION TEST RESULTS

4.1	Conducted disturbance voltage - Mains	VERDICT: PASS
-----	---------------------------------------	---------------

Standard	EN IEC 55014-1
Basic standard	EN 55016-2-1

Limits

Frequency range [MHz]	Limit: QP [dB(μV) ¹⁾	Limit: AV [dB(μV) ¹⁾	IF BW	Detector(s)
0,15 - 0,50	66 - 56 ²⁾	59 - 46 ²⁾	9 KHz	QP, CAV
0,50 - 5,0	56	46	9 KHz	QP, CAV
5,0 - 30	60	50	9 KHz	QP, CAV

¹⁾ At the transition frequency, the lower limit applies.

²⁾ The limit decreases linearly with the logarithm of the frequency.

Performed measurements

Scan range (0.9 – 1.1 U _N)	☐	198 – 264 V _{AC}	☐	207 – 253 V _{AC}			☒	230 V _{AC}		
Tested terminal(s) / port	☒	AC mains input power	☒	N	☒	L1	☐	L2	☐	L3
	☐	DC mains input power	☐	Positive (+)			☐	Negative (-)		
Voltage – Mains [V]	230 Vac									
Frequency – Mains [Hz]	50 Hz									
Test method applied	☒	Artificial mains network								
	☐	Voltage probe								
Test setup	☐	Table top		☐	Artificial hand applied					
	☒	Floor standing		☐	Other:					
	Refer to the Annex 3 for test setup photo(s).									
Operating mode(s) used	Mode 1, Mode 2									
Remark	---									

See next page.

Measurement data	Port under test	AC mains power input
Operating mode / voltage / frequency used during the test		Mode 2 / 230 Vac / 50 Hz

Line:



EMI Final Results

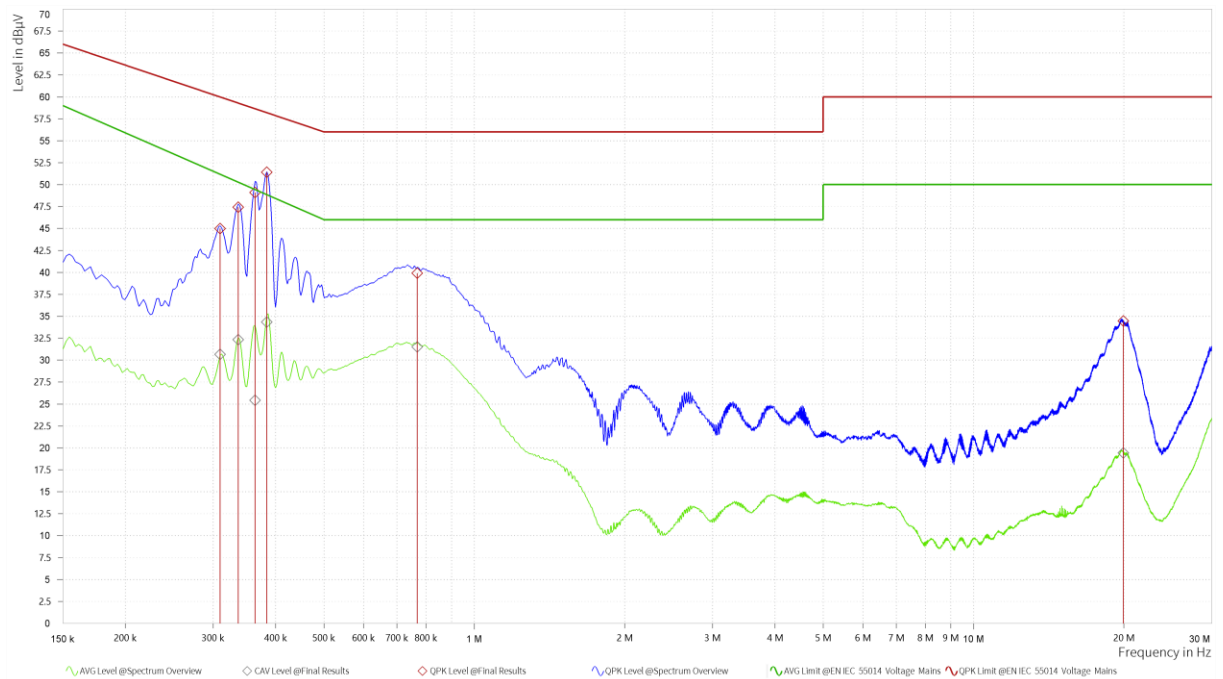
Frequency [MHz]	QPK Level [dBµV]	QPK Limit [dBµV]	QPK Margin [dB]	CAV Level [dBµV]	CAV: AVG Limit [dBµV]	CAV Margin [dB]	Correction [dB]	Line
0.310	44.13	59.98	15.85	29.58	51.17	21.59	9.68	L1
0.337	46.51	59.28	12.77	29.26	50.27	21.01	9.69	L1
0.364	49.20	58.64	9.44	24.35	49.44	25.09	9.69	L1
0.384	50.61	58.19	7.58	30.00	48.85	18.85	9.70	L1
0.796	39.51	56.00	16.49	31.01	46.00	14.99	9.68	L1
20.063	34.38	60.00	25.62	19.21	50.00	30.79	9.94	L1

1. QPK Level [dBµV]=Reading QPK Level [dBµV] + Correction [dB]
2. CAV Level [dBµV]=Reading CAV Level [dBµV] + Correction [dB]
3. Correction [dB]= AMN VDF [dB] + Cable Loss [dB]

Remark Pre-test mode 1-2 during the testing, then listed the worst data (Mode 2) in this report.

Measurement data	Port under test	AC mains power input
Operating mode / voltage / frequency used during the test		Mode 2 / 230 Vac / 50 Hz

Neutral



EMI Final Results

Frequency [MHz]	QPK Level [dBµV]	QPK Limit [dBµV]	QPK Margin [dB]	CAV Level [dBµV]	CAV: AVG Limit [dBµV]	CAV Margin [dB]	Correction [dB]	Line
0.310	44.98	59.98	15.00	30.66	51.17	20.51	9.71	N
0.337	47.41	59.28	11.87	32.31	50.27	17.96	9.72	N
0.364	49.09	58.64	9.56	25.40	49.44	24.03	9.72	N
0.384	51.41	58.19	6.78	34.34	48.85	14.51	9.73	N
0.769	39.90	56.00	16.10	31.49	46.00	14.51	9.60	N
19.950	34.45	60.00	25.55	19.41	50.00	30.59	9.95	N

1. QPK Level [dBµV]=Reading QPK Level [dBµV] + Correction [dB]
2. CAV Level [dBµV]=Reading CAV Level [dBµV] + Correction [dB]
3. Correction [dB]= AMN VDF [dB] + Cable Loss [dB]

Remark	Pre-test mode 1-2 during the testing, then listed the worst data (Mode 2) in this report.
--------	---

4.2 Disturbance power (30 MHz – 300 MHz)	VERDICT: PASS
---	----------------------

Standard	EN IEC 55014-1
Basic standard	EN 55016-2-2

Limits

Frequency range [MHz]	Limit: QP [dB(pW)]	Limit: AV [dB(pW)]	IF BW	Detector(s)
30 - 300	45 – 55 ¹⁾	35 – 45 ¹⁾	120 KHz	QP, CAV
Margin				
200 - 300	0 – 10 ¹⁾	---	120 KHz	QP, CAV
¹⁾ The limit increases linearly with the frequency.				

Performed measurements

Port(s) under test						
☒	AC input power	☐	Load	☐	Control	
☒	Other: DC output power	☐	Other:	☐	Other:	
Voltage – Mains [V]		230 Vac				
Frequency – Mains [Hz]		50 Hz				
Test setup		☐	Table top	☒	Floor standing	
		☐	Other:			
		Refer to the Annex 3 for test setup photo(s).				
Conditions for exemption from measurements above 300 MHz		☒	“Limits” reduced by “Over Limit” applied and passed			
		☒	Maximum clock frequency < 30 MHz			
Operating mode(s) used		Mode 1, Mode 2				
Remark		---				

See next page.

Measurement data	Port under test	AC input power
Operating mode / voltage / frequency used during the test		Mode 2 / 230 Vac / 50 Hz

Level (dBpW)
Date: 2024-12-23

Item	Freq. MHz	Reading Level dBpW	C.F dB	Emission Level dBpW	Limit dBpW	Over Limit dB	Detector
1	41.34	15.61	26.90	42.51	45.43	-2.92	QP
2	41.34	0.21	26.90	27.11	35.43	-8.32	Average
3	78.60	3.82	26.38	30.20	46.81	-16.61	QP
4	78.60	-5.58	26.38	20.80	36.81	-16.01	Average
5	113.97	6.81	23.95	30.76	48.12	-17.36	QP
6	113.97	-9.49	23.95	14.46	38.12	-23.66	Average
7	120.72	5.21	24.37	29.58	48.37	-18.79	QP
8	120.72	-11.79	24.37	12.58	38.37	-25.79	Average
9	206.58	3.11	22.15	25.26	50.89	-25.63	QP
10	206.58	-13.69	22.15	8.46	41.55	-33.09	Average
11	224.40	6.82	22.23	29.05	49.77	-20.72	QP
12	224.40	-10.98	22.23	11.25	42.21	-30.96	Average

Remarks : 1. C.F(Correction Factor)=Absorbing Clamp factor+Cable Loss.
2. Emission Level= Read Level+ C.F(Correction Factor)

4.3	Discontinuous disturbance (clicks) on AC power leads	VERDICT: PASS
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Standard	EN IEC 55014-1, CISPR 14-1		
Frequency [MHz]	Limit: QP [dB(μV)]	IF BW	Detector
0.15	66	9 KHz	Quasi-Peak (QP)
0.50	56	9 KHz	Quasi-Peak (QP)
1.40	56	9 KHz	Quasi-Peak (QP)
30.0	60	9 KHz	Quasi-Peak (QP)

Performed measurements

Voltage – Mains [V]	230 Vac		
Frequency – Mains [Hz]	50 Hz		
Test method applied	☒	Artificial mains network	
	☐	Voltage probe	
Test setup	☐	Table top	☒ Floor standing
	☐	Other: Hand-held equipment	
	Refer to the Annex 3 for test setup photo(s).		
Operating mode(s) used	Mode 2		
Remark	---		

Measurement results		☒	Neutral	☒	Line 1	☐	Line 2	☐	Line 3
Frequency (MHz)	First Measurement: Determination of the limit L_q – Quasi-peak								
	Limit L (dBμV)	Number of short clicks	Number of long clicks	Number of clicks – N_1	Time of meas. (min.)	Click rate N	Increased limit (dB)	Increased Limit L_q	
0.15	66	0	0	0	120	0	44	110	
0.50	56	0	0	0	120	0	44	100	
1.40	56	0	0	0	120	0	44	100	
30.0	60	0	0	0	120	0	44	104	
☐	The calculated click rate N is not more than 5 times per minute and all the clicks are classified as short ($t \leq 10$ ms). Thus, the EUT is deemed to comply with the limits without any further measurement at an increased limit.								
Frequency (MHz)	Second measurement with Limit = L_q (Upper quartile method):								
	Limit L_q (dBμV)	Number of clicks – N_2	Number of authorized clicks $N_2 \leq N_1/4$				Verdict		
0.15									
0.50									
1.40									
30.0									
Remark	The amplitudes of the observed disturbances were all below the limit for continuous disturbance, these are not considered to be clicks.								

4.4	Harmonic current emissions	VERDICT: PASS
-----	-----------------------------------	----------------------

Standard	EN IEC 61000-3-2	
Exclusions (For these categories of equipment, limits are not specified in the EN IEC 61000-3-2 standard)	☐	Arc welding equipment intended for professional use.
	☐	System(s) with nominal voltage(s) less than 220 V _{AC} (line-to-neutral).
	☐	Equipment with rated power of ≤ 75 W (other than lighting equipment).
	☐	Professional equipment with total rated power > 1 kW.
	☐	Symmetrically controlled heating elements with a rated power ≥ 200 W.
	☐	Independent dimmers for incandescent lamps with rated power ≤ 1 kW.

Classification		
☒	Class A	All apparatus not classified as Class B, C or D
☐	Class B	Portable tools
☐	Class C	☐ Lighting equipment with active input power > 25 W
		☐ Lighting equipment with active input power ≤ 25 W (First requirement, Table 3 column 2)
		☐ Lighting equipment with active input power ≤ 25 W (Second requirement)
☐	Class D	Personal computers, television receivers

Performed measurements

Port under test	AC mains power input					
Voltage – Mains [V]	230 Vac					
Frequency – Mains [Hz]	50 Hz					
Observation peroid	☐	6.5 min.	☒	2.5 min.	☐	Other:
Version of measurement instrument standard used EN / IEC61000-4-7 (Cl. 7)	☒	EN 61000-4-7:2002 + AM1:2009 (IEC 61000-4-7:2002+AM1:2008)				
	☐	EN 61000-4-7:1991				
Control principle used in the EUT	☒	Comply with the requirements of the Clause 6.1 (EN IEC / IEC 61000-3-2).				
	☐	Not comply with the requirements of the Clause 6.1 (EN IEC / IEC 61000-3-2).				
Operating mode(s) used	Mode 1, Mode 2					
Remark	1) Mode 1 of EUT is regarded as an “Equipment with rated power of ≤ 75 W”. According to “Clause 7, Figure 1 - Flowchart for determining conformity” the EUT is deemed to comply with the requirements of the EN 61000-3-2 standard.					

See next page.

Measurement data	Port under test	AC input power	
Operating mode / voltage / frequency used during the test		Mode 2 / 230 Vac / 50 Hz	
Measure Results			
Standard Specific Results for IEC 61000-3-2 (Edition 5.1)			
Standard Group:	Industry		
Standard Name:	IEC 61000-3-2 (Edition 5.1)		
	Limits for harmonic current emissions (equipment input current < 16 A per phase)		
Device Under Test:	PASS		
Power Source:	PASS		
Connection Type:	L - N		
Main Line:	230 V, 50 Hz		
Classification:	Class A		
Appli. of Limits:	less than or equal to 150 % (Without POHC Enhancement)		
Grouping:	not applied		
Evaluation of Partial Odd Harmonics Enhancement			
Partial Odd Harmonic Currents			
	measured value	limit	result
Line 1:	0.032 A (1.15 %)	0.251 A	PASS
Evaluation for current limits			
	without enhancement	with enhancement	need enhancement
Line 1:	PASS	PASS	No
Enhancement is not applied			
Check Harmonics 2..40			
First detected harmonic order > 150 %			
Line 1:	None		
Harmonics orders > 150 %			
Line 1:	None		
Harmonics orders with average > 100 %			
Line 1:	None		
Measured values			
Fundamental Current			
Line 1:	2.72 A		
Active input Power			
Line 1:	618.46 W *		
Circuit power factor			
Line 1:	0.933 *		
*Absolute value.			

Measurement data				Port under test		AC input power			
Current Test Result									
Average and Maximum harmonic current results									
Hn	Average				Maximum				Harmonic Result
	I _{eff} [A]	of Limit [%]	Limit [A]	Result	I _{eff} [A]	of Limit [%]	Limit [A]	Result	
1	0.352				2.723				
2	0.001	0.106	1.080	n/a	0.015	0.911	1.620	n/a	PASS
3	0.137	5.963	2.300	PASS	0.838	24.292	3.450	PASS	PASS
4	0.001	0.194	0.430	n/a	0.011	1.687	0.645	n/a	PASS
5	0.086	7.576	1.140	PASS	0.356	20.818	1.710	PASS	PASS
6	0.001	0.313	0.300	n/a	0.008	1.765	0.450	n/a	PASS
7	0.072	9.343	0.770	PASS	0.157	13.595	1.155	PASS	PASS
8	0.001	0.303	0.230	n/a	0.006	1.740	0.345	n/a	PASS
9	0.060	14.886	0.400	PASS	0.076	12.705	0.600	PASS	PASS
10	0.001	0.523	0.184	n/a	0.005	1.785	0.276	n/a	PASS
11	0.048	14.623	0.330	PASS	0.058	11.620	0.495	PASS	PASS
12	0.001	0.408	0.153	n/a	0.004	1.561	0.230	n/a	PASS
13	0.038	18.015	0.210	PASS	0.046	14.598	0.315	PASS	PASS
14	0.001	0.571	0.131	n/a	0.003	1.444	0.197	n/a	PASS
15	0.029	19.066	0.150	PASS	0.035	15.671	0.225	PASS	PASS
16	0.001	0.538	0.115	n/a	0.002	1.309	0.173	n/a	PASS
17	0.021	15.942	0.132	PASS	0.026	13.321	0.199	PASS	PASS
18	0.000	0.424	0.102	n/a	0.002	1.325	0.153	n/a	PASS
19	0.015	13.048	0.118	n/a	0.019	10.915	0.178	PASS	PASS
20	0.001	0.626	0.092	n/a	0.003	1.828	0.138	n/a	PASS
21	0.014	13.077	0.107	n/a	0.018	10.893	0.161	PASS	PASS
22	0.000	0.460	0.084	n/a	0.002	1.656	0.125	n/a	PASS
23	0.013	13.175	0.098	n/a	0.016	10.803	0.147	n/a	PASS
24	0.001	0.686	0.077	n/a	0.002	1.871	0.115	n/a	PASS
25	0.013	14.495	0.090	n/a	0.016	11.751	0.135	n/a	PASS
26	0.000	0.706	0.071	n/a	0.002	1.615	0.106	n/a	PASS
27	0.012	14.521	0.083	n/a	0.015	11.850	0.125	n/a	PASS
28	0.000	0.590	0.066	n/a	0.001	1.505	0.099	n/a	PASS
29	0.011	13.893	0.078	n/a	0.014	11.605	0.116	n/a	PASS
30	0.001	0.930	0.061	n/a	0.001	1.530	0.092	n/a	PASS
31	0.010	13.378	0.073	n/a	0.012	11.334	0.109	n/a	PASS
32	0.000	0.548	0.058	n/a	0.001	1.548	0.086	n/a	PASS
33	0.007	10.665	0.068	n/a	0.009	9.162	0.102	n/a	PASS
34	0.000	0.755	0.054	n/a	0.001	1.348	0.081	n/a	PASS
35	0.006	9.709	0.064	n/a	0.008	8.348	0.096	n/a	PASS
36	0.000	0.725	0.051	n/a	0.001	1.284	0.077	n/a	PASS
37	0.005	8.594	0.061	n/a	0.007	7.188	0.091	n/a	PASS
38	0.000	0.537	0.048	n/a	0.001	1.765	0.073	n/a	PASS
39	0.005	8.472	0.058	n/a	0.006	6.968	0.087	n/a	PASS
40	0.000	0.940	0.046	n/a	0.001	2.059	0.069	n/a	PASS
Note: Harmonic currents less than 0.6 % of the input current measured under the test conditions, or less than 5 mA, whichever is greater, are disregarded.									

4.5	Voltage changes, voltage fluctuations and flicker	VERDICT: PASS
-----	--	----------------------

Standard	EN 61000-3-3
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Limits

P _{ST} (Short term flicker)	☒	≤ 1	☐	Not Applicable
P _{LT} (Long term flicker)	☒	≤ 0.65 ¹⁾	☒	Not Applicable
d _C (Relative Voltage change)	☒	≤ 3.3%	☐	Not Applicable
T _{MAX} (Maximum time duration)	☒	≤ 500ms	☐	Not Applicable
d _{MAX} (Max. voltage change)	☒	≤ 4% ¹⁾	☒	6%
	☐	7%	☐	Not Applicable
<u>Supplemental information:</u>				

Performed measurements

Reason for not performing the measurement(s)	☐	Tests are not necessary because the EUT is unlikely to produce significant voltage fluctuations or flicker (clause 6.1).				
Port under test	AC Mains power input					
Voltage – Mains [V]	230 Vac					
Frequency – Mains [Hz]	50 Hz					
Test method	☒	Flickermeter according EN / IEC 61000-4-15:2011				
	☐	Simulation (Clause 4.2.3 of EN / IEC 61000-3-3)				
	☐	Analytical method (Clause 4.2.4 of EN / IEC 61000-3-3)				
	☐	Use of $P_{st} = 1$ curve (Clause 4.2.5 of EN / IEC 61000-3-3)				
Observation peroid	☒	10 min.	☒	120 min. ¹⁾	☐	Other:
	☐	24 times switching according to Annex B				
Operating mode(s) used	Mode 1, Mode 2					
Remark	1) For Mode 1					

See next page.

Measurement data		Port under test	AC input power
Operating mode used during the test		230 Vac / 50 Hz	
Results for Mode 1			
Tmax (dt > 3,3%)		0.0 ms	
Maximum relative voltage change d _{MAX}		0.802 %	
Relative Voltage change d _c		0.683%	
Short term flicker P _{ST}		0.166	
Long term flicker P _{LT}		0.076	
Results for Mode 2			
Tmax (dt > 3,3%)		0.0 ms	
Maximum relative voltage change d _{MAX}		0.784 %	
Relative Voltage change d _c		0.665 %	
Short term flicker P _{ST}		0.163	
Long term flicker P _{LT}		---	

5 IMMUNITY TEST RESULTS

5.1 Performance (Compliance) criteria

[According to EN IEC 55014-2 (CISPR 14-2)]

Performance criteria A : The apparatus shall continue to operate as intended during the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer when the apparatus is used as intended. If the minimum performance level or the permissible performance loss is not specified by the manufacturer then either of these may be derived from the product description and documentation and from what the user may reasonably expect from the apparatus if used as intended.

Performance criteria B : The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer when the apparatus is used as intended. During the test, degradation of performance is allowed however no change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer then either of these may be derived from the product description and documentation and from what the user may reasonable expect from the apparatus if used as intended.

Performance criteria C : Temporary loss of function is allowed provided the function is self- recoverable or can be restored by the operation of the controls or by any operation specified in the instruction for use.

5.1.1 Performance criteria related to immunity tests

Immunity test	Performance criteria
Electrostatic discharge	B
Radio-frequency electromagnetic fields	A
Fast transients	B
Surge transient	B
Injected currents (radio-frequency common mode)	A
Voltage dips and short interruptions	C

5.1.2 Manufacturer defined performance criteria

Not provided.

5.2 Monitored – Checked Functions / Parameters

During the immunity tests the following functions of the EUT has/have been monitored/checked.

☐	Motor speed	☐	Display data
☐	Switching	☐	Data storage
☐	Standby mode	☐	Sensor functions
☐	Temperature	☐	Audible signals
☐	Power consumption	☒	Others : Functional status
☐	AC mains input current	☐	Others :
☐	Timing	☐	Others :
☐	Illumination	☐	Others :
<u>Supplementary information :</u>			

Immunity test	Monitored - Checked function(s)/parameter(s) during / after the test	Method
Electrostatic discharge	Functional status	Visual
Radio-frequency electromagnetic fields	---	Visual / Camera
Fast transients	Functional status	Visual
Surge transient	Functional status	Visual
Injected currents (radio-frequency common mode)	Functional status	Visual
Voltage dips and short interruptions	Functional status	Visual
<u>Supplementary information :</u>		

5.3 Electrostatic discharge immunity

VERDICT: PASS

Electrostatic discharges (ESD) are the result of persons or objects that accumulate static electricity due to for instance walking on synthetic carpets. The ESD can influence the operation of equipment or damage its electronics, either by a direct discharge or indirectly by coupling or radiation. Both effects are simulated during the tests.

Requirements

Standard	EN IEC 55014-2							
Basic standard	EN 61000-4-2							
Port under test	Enclosure							
Air discharges ¹⁾	☐	±2 kV	☐	±4 kV	☒	±8 kV	☐	kV
Contact discharges ¹⁾	☐	±2 kV	☒	±4 kV	☐	±8 kV	☐	kV
Number of discharges	≥ 10 per polarity with ≥ 1 sec interval.							
¹⁾ Tests with lower voltages are not required.								

Performed tests for test model

Set-up	☐	Table-top	☒	Floor standing
Ambient temperature [°C]	24 °C		Relative Humidity air [%]	48 %
Voltage – Mains [V]	230 Vac			
Frequency – Mains [Hz]	50 Hz			
Operating mode(s) used	Mode 1, Mode 2			

Test Point		Test Voltage [kV] & Polarity	Coupling type	# of applied discharges / polarity	Discharge interval [s]
☒	Points on conductive surface.	±4	Contact	10	1
☒	Points on non-conductive surface.	±8	Air	10	1
☐	HCP top side.	±4	Contact	10	1
☐	HCP bottom side.	±4	Contact	10	1
☒	VCP right side.	±4	Contact	10	1
☒	VCP left side.	±4	Contact	10	1
☒	VCP front side.	±4	Contact	10	1
☒	VCP rear side.	±4	Contact	10	1
Observation(s)		During the test no loss of performance was observed. After the test the EUT functioned as intended. No unacceptable loss of performance or data was observed.			
Supplementary information:					

5.4	Electrical Fast Transients immunity	VERDICT: PASS
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The EFT immunity test simulates disturbances by bursts of very short transients caused for example by switching off loads such as an AC motor or bouncing relay contacts. The transients are likely to disturb electronics but less likely to cause damage.

Requirements

Standard	EN IEC 55014-2			
Basic standard	EN 61000-4-4			
Pulse characteristics	5/50 ns			
Port		Test level	Repetition frequency	Duration
☒	AC input-output power ¹⁾	± 1000 V	5 KHz	2 min. / polarity
☐	DC input-output power ²⁾	± 500 V	5 KHz	2 min. / polarity
☐	Signal and Control lines ³⁾	± 500 V	5 KHz	2 min. / polarity
¹⁾ For extra low voltage a.c ports, this testing is only applicable to ports interfacing with cables whose total length may exceed 3 m according to the manufacturer's functional specification. ²⁾ Not applicable to battery operated appliances that cannot be connected to the mains while in use. ³⁾ Applicable only to ports interfacing with cables whose total length may exceed 3 m according to the manufacturer's functional specification.				

Performed tests for test model

Voltage – Mains [V]	230 Vac			
Frequency – Mains [Hz]	50 Hz			
Operating mode(s) used	Mode 1, Mode 2			
Test Set-up (see annex 3 for photo)	☒	Equipment standing on floor at (0,1 ± 0,01) m above ground plane		
	☐	Equipment on the table (0,1 ± 0,01) m above ground plane		
	☐	Artificial hand applied. Location refer to annex 3.		
Coupling	☒	Common mode	☐	Other:

Port(s) under test	Test Voltage & Polarity	Repetition Frequency	Test duration / polarity	Injection method			
AC mains power input	± 1000 V	5 KHz	2 min	☒	CDN	☐	Clamp
Observation(s)	During the test no loss of performance was observed. After the test the EUT functioned as intended. No unacceptable loss of performance or data was observed.						

5.5	Surge transient immunity	VERDICT: PASS
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The surge transient immunity test simulates the surges that are caused by over-voltages due to indirect (induced) lightning transients. The pulse is a slow transient with high-energy contents and due to its long duration may cause damage to an unprotected EUT.

Requirements

Standard	EN IEC 55014-2		
Basic standard	EN 61000-4-5		
Pulse characteristics	1,2/50 μ s Voltage; 8/20 μ s Current		
Repetition rate	$\leq$ 60 secs. (for each test level and phase angle)		
Number of pulses	5 pulses (at each polarity and phase angle)		
Port	Test level & Polarity & Coupling		Phase angle [°]
	Line to Line	Line to Earth	
AC input power ¹⁾	+ 1 kV	+ 2 kV	90
AC input power ¹⁾	- 1 kV	- 2 kV	270
¹⁾ Tests with lower voltages are not required.			

Performed tests for test model

Voltage – Mains [V]	230 Vac
Frequency – Mains [Hz]	50 Hz
Operating mode(s) used	Mode 1, Mode 2
Repetition rate	60 secs. (for each test level and phase angle)
Number of pulses	5 pulses (at each polarity and phase angle)

Port(s) under test		Coupling	Test level & Polarity	Phase angle [°]	Remark
☒	AC mains input power	Line to Neutral	+1 kV	90	---
☒	AC mains input power	Line to Neutral	-1 kV	270	---
☐	AC mains input power	Line to Earth	+2 kV	90	---
☐	AC mains input power	Line to Earth	-2 kV	270	---
☐	AC mains input power	Neutral to Earth	+2 kV	90	---
☐	AC mains input power	Neutral to Earth	-2 kV	270	---
Observation(s)		During the test no loss of performance was observed. After the test the EUT functioned as intended. No unacceptable loss of performance or data was observed.			
Supplementary information:					

5.6	Injected currents (RF common mode) immunity	VERDICT: PASS
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During this test the immunity of the equipment for induced or conducted electromagnetic fields is checked. Fields generated by radio and other transmitters cause RF voltages in long cables like the mains network. This test reproduces these induced disturbing voltages by injecting them to the EUT via the cabling.

Requirements

Standard		EN IEC 55014-2		
Basic standard		EN 61000-4-6		
Frequency range		Modulation	Step size	Dwell time
☐	0,15 – 80 MHz	80% AM (1kHz)	≤ 1%	≥ 0,5 s
☒	0,15 – 230 MHz	80% AM (1kHz)	≤ 1%	≥ 0,5 s
Port		Test level, U ₀		
☒	AC input-output power ¹⁾	3 V		
☐	DC input-output power ^{2) 3)}	1 V		
☐	Signal and Control lines ⁴⁾	1 V		
¹⁾ For extra low voltage a.c ports, this testing is only applicable to ports interfacing with cables whose total length may exceed 3 m according to the manufacturer's functional specification.				
²⁾ Not applicable to battery operated appliances that cannot be connected to the mains while in use.				
³⁾ Applicable to battery operated appliances that can be connected to the mains while in use, or to appliances for which the length of d.c. cables may exceed 3 m according to the manufacturer's functional specification.				
⁴⁾ Applicable only to ports interfacing with cables whose total length may exceed 3 m according to the manufacturer's functional specification.				

Performed tests for test model

Frequency range (applied)	Modulation (applied)	Step size (applied)
☐ 0,15 – 80 MHz	☒ 0,15 – 230 MHz	80% AM (1kHz)
Voltage – Mains [V]	230 Vac	Frequency – Mains [Hz]
Operating mode(s) used	Mode 1, Mode 2	50 Hz
Test set-up (see annex 3 for photo)	☒ Equipment standing on floor at $(0,1 \pm 0,01)$ m above ground plane. ☐ Equipment on the table $(0,1 \pm 0,01)$ m above ground plane. ☐ Artificial hand applied. Location refer to annex 3.	

Port(s) under test		Test Level (applied)	Injection method	Dwell time (applied)	Remark
AC mains power input		3 V	CDN-M2	3 s	---
Observation(s)	During the test no loss of performance was observed. After the test the EUT functioned as intended. No unacceptable loss of performance was observed.				
<u>Supplementary information:</u>					

5.7	Power supply interruptions and dips immunity	VERDICT: PASS
-----	---	----------------------

The purpose of the test is to verify the immunity of the equipment against voltage dips and voltage interruptions. It helps to ensure that the equipment functions properly (as expected and safely) with power supply fluctuations. Voltage dips and interruptions are caused by faults in the LV, MV, HV networks (short-circuit or ground faults).

Requirements

Standard	EN IEC 55014-2			
Basic standard	EN 61000-4-11			
# of dips & interruptions	3 dips / interruptions for each test level and phase angle			
Interval between events	≥ 10 seconds			
Port	Test level ¹⁾	Period (Cycles)		Performance Criterion
		50 Hz	60 Hz	
AC input power port	$U_{NOM} - 100\%$	0,5	0,5	C; Refer to the chapter 5.1 for details.
AC input power port	$U_{NOM} - 60\%$	10	12	C; Refer to the chapter 5.1 for details.
AC input power port	$U_{NOM} - 30\%$	25	30	C; Refer to the chapter 5.1 for details.
¹⁾ Changes to the voltage level shall occur at a zero crossing point in the a.c. voltage waveform. NOTE: Where the equipment has a rated voltage range the following shall apply: - If the voltage range does not exceed 20% of the lower voltage specified for the rated voltage range. A single voltage within that range may be selected for testing. - In all other cases, the test procedure shall be applied for both the lowest and highest voltages declared in the voltage range.				

Performed tests for test model

U _{NOM} [V _{AC}]	Terminal	Voltage dip [% U _{NOM}]	Duration [cycles]		Repetition rate [s]	Number of dips per test	Phase angle [°]
			50 Hz	60 Hz			
230	L-N	0	0,5	0,5	10	3	0, 180
230	L-N	40	10	12	10	3	0, 180
230	L-N	70	25	30	10	3	0, 180
Operating mode(s) used		Mode 1, Mode 2					
Observation(s)		During the test no loss of performance was observed. After the test the EUT functioned as intended. No unacceptable loss of performance was observed.					
<u>Supplementary information:</u>							

6 IDENTIFICATION OF THE EQUIPMENT UNDER TEST

EUT PHOTOS



7 ANNEX 1 - MEASUREMENT UNCERTAINTIES

The table(s) below show(s) measurement uncertainties of the EMC test set-ups. The reported expanded uncertainties are based on a standard uncertainty multiplied by a coverage factor of $k=2$, providing a level of confidence of approximately 95%.

Emission tests		U_{LAB}	U_{CISPR}
Conducted disturbance using AMN 150 kHz – 30 MHz	Shielded Room No.1 AMN: R&S ENV216	2.6 dB	3.4 dB
Disturbance power using Absorbing clamp 30 MHz – 300 MHz	Absorbing clamp: MDS 21B	4.0 dB	4.5 dB
Harmonic current emissions		0.6 %	N/A
Voltage changes, Voltage fluctuations and Flicker		2.8 %	N/A

Immunity tests		U_{LAB}	U_{IEC} (Recommendation)
Electrostatic discharge	Rise time (t_r)	12.4 %	≤ 15 %
	Peak current (I_p)	6.4 %	≤ 7 %
	Current at 30 ns (I_{30})	6.4 %	≤ 7 %
	Current at 60 ns (I_{60})	6.4 %	≤ 7 %
Electrical fast transients/burst	Rise time (t_r)	7.2 %	N/A
	Peak voltage (V_p)	6.8 %	
	Pulse width (t_w)	5.4 %	
Surge	Front time (t_f)	4.3 %	N/A
	Peak voltage (V_p)	6.7 %	
	Voltage duration (T_d)	1.3 %	
Conducted disturbances, induced by radio-frequency fields	CDN (v)	1.6 dB	N/A
Voltage dips, short interruptions and voltage variations	Time at reduced voltage (t_s)	3.0 %	N/A
	Voltage rising time (t_r)	3.0 %	
	Voltage fall time (t_f)	3.0 %	
	Voltage dips (v)	0.4 %	

8 ANNEX 2 – USED EQUIPMENT

Conducted disturbance -Shielded Room No.1

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
EMI test receiver	R&S	ESR3	102958	2025/06/13
Artificial Mains Network	R&S	ENV216	102772	2025/06/13
Software	R&S	ELEKTRA	4.32.0	N/A

Disturbance power

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
EMI test receiver	R&S	ESCI	101351	2025/06/13
EMI absorbing clamp	SCHWARZBECK	MDS 21B	4183	2025/06/24
Software	AUDIX	e3	V8.130528a	N/A

Discontinuous disturbance (clicks)

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
EMI test receiver	R&S	ESCI	101351	2025/06/13
Artificial Mains Network	SCHWARZBECK	MDS 21B	4183	2025/06/24
Software	R&S	Click Rate Analyzer	2.52	N/A

Harmonic current emissions & Voltage changes, voltage fluctuations and flicker

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
Harmonic currents and flicker tester	EMTEST	DPA 500N	P2114250803	2026/01/20
AC power source	EMTEST	Netwave7-400	P2136256253	2026/01/20
Software	AMETEK	NET.CONTROL	3.1.3	N/A

Electrostatic discharge immunity

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
ESD generator	TESEQ	NSG 435	6716	2025/08/22

Fast transient immunity

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
Compact immunity test system	TESEQ	NSG-3040-MF	2006/EFT:0535	2025/06/13

Surge immunity

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
Compact immunity test system	TESEQ	NSG-3040-MF	2006 /SURGE:1234	2025/06/13

Injected currents immunity

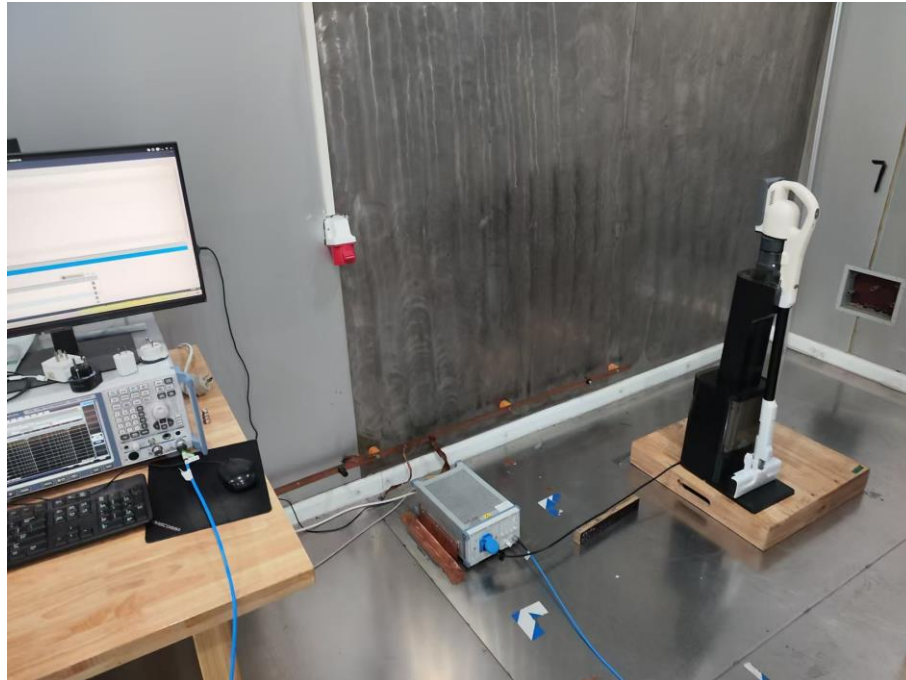
Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
Compact immunity test system	TESEQ	NSG 4070-30	35895	2025/06/13
Coupling/Decoupling Network (CDN)	TESEQ	CDN M016S	34640	2025/06/13
Attenuator	TESEQ	ANT 6050	34847	2025/06/13

Voltage dips and short interruptions immunity

Equipment	Manufacturer	Model No.	Serial No.	Cal. due date
Compact immunity test system	TESEQ	NSG-3040-MF	2006/DIPS:2062	2025/06/13
Automatic step transformer with circuit breaker	TESEQ	INA 6502-CIB	217	2025/06/13

9 ANNEX 3 - TEST PHOTOS

Conducted disturbance voltage at mains terminals



Disturbance power



Harmonic current emissions & Voltage changes, voltage fluctuations and flicker



Electrostatic discharge immunity



Fast transients & Voltage dips and short interruptions & Surge



Injected currents (radio-frequency common mode)



End of the report